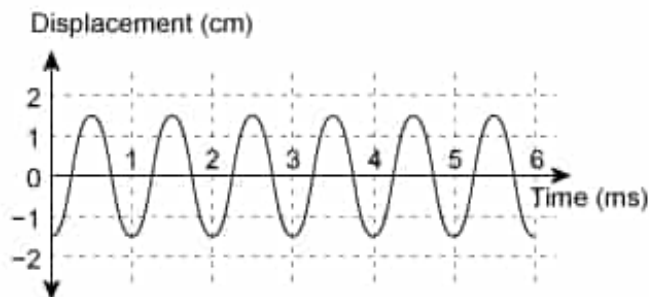
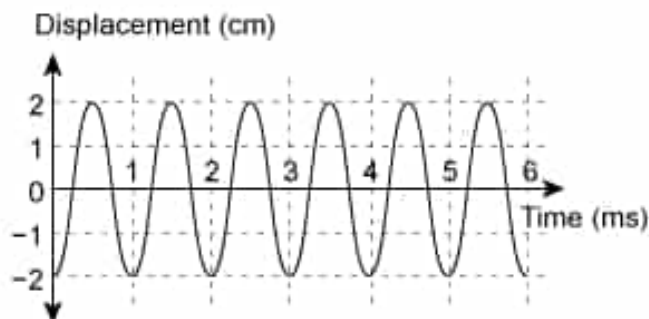


The speed of waves on a piano string is directly proportional to the square root of the string tension. A piano tuner presses a key on the piano and uses an oscilloscope to visualize the resulting sound wave. The oscilloscope displays a graph of displacement as a function of time for the note.



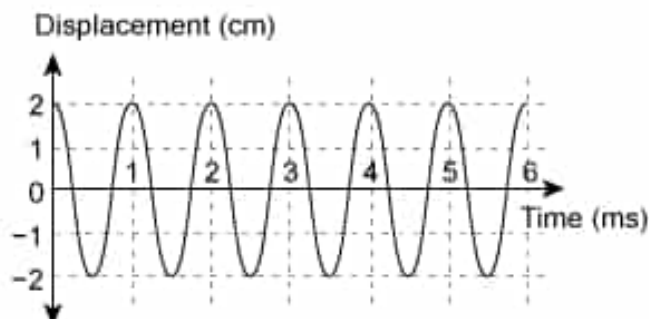
The tuner tightens the string and presses the same key on the piano. Which of the following could show the corresponding graph for the new sound wave?

☐ A.



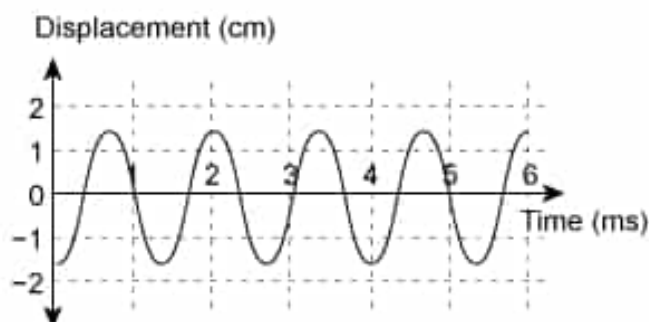
[15%]

☐ B.



[5%]

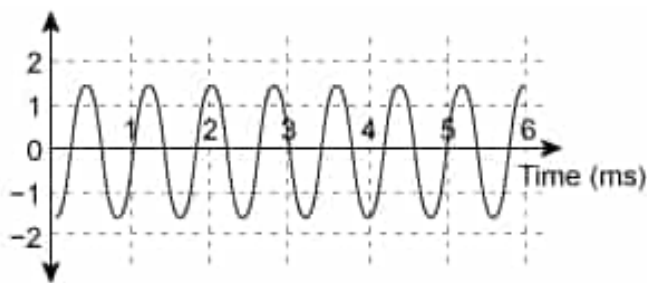
☐ C.



[7%]

☒ D.

Displacement (cm)



[71%]

Correct

71% answered correctly

Explanation:

Increasing string's tension increases frequency

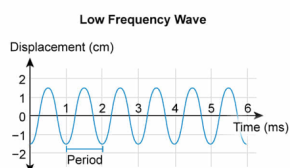
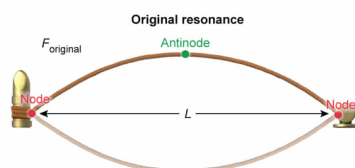
$$f = \frac{v}{\lambda}$$

$$\text{Period} = \frac{1}{f}$$

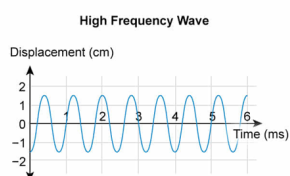
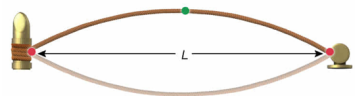
$$v = \sqrt{\frac{F}{\mu}}$$

$$f = \frac{1}{\lambda} \sqrt{\frac{F}{\mu}}$$

- F String tension
- f Frequency
- L Length
- v Speed of waves
- μ String's linear density
- λ Wavelength



New resonance (string has higher tension)
 $F_{\text{new}} > F_{\text{original}} \rightarrow v_{\text{new}} > v_{\text{original}} \rightarrow f_{\text{new}} > f_{\text{original}}$



The **frequency** f of the wave is the ratio of the wave speed v to the **wavelength** λ :

$$f = \frac{v}{\lambda}$$

Furthermore, the speed of a wave on a string is equal to the square root of the ratio of the string tension force F and linear density μ of the string:

$$v = \sqrt{\frac{F}{\mu}}$$

Hence, wave frequency is **directly proportional** to the square root of the tension force:

$$f = \frac{1}{\lambda} \cdot \sqrt{\frac{F}{\mu}}$$

For constant wavelength and density, if the string tension increases, then frequency also increases.

In this question, the piano string has a fixed length and when the tuner presses the key, the string vibrates with a constant wavelength. However, as the tuner increases the tension in the string, the wave frequency increases and the period $T = 1/f$ of the wave decreases.

Therefore, the graph of displacement as a function of time must show crests that are closer together in time, consistent with the increased frequency and decreased period of answer choice D.

(Choice A) This graph shows a sound wave with higher amplitude.

However, the peaks are the same distance apart, which indicates the frequency remained the same.

(Choice B) This graph shows a sound wave shifted to the right with higher amplitude. However, the peaks are the same distance apart, not closer together.

(Choice C) This graph shows a sound wave with lower amplitude and peaks farther apart, which represents a sound wave with a longer period and lower frequency.

Educational objective:

For constant wavelength, an increasing the string tension also increases wave speed and frequency. Higher frequency results in crests that are closer together on a graph of displacement as a function of time.

Time spent:0 sec

Subject:
Physics

QID:404146

System:
Light and Sound

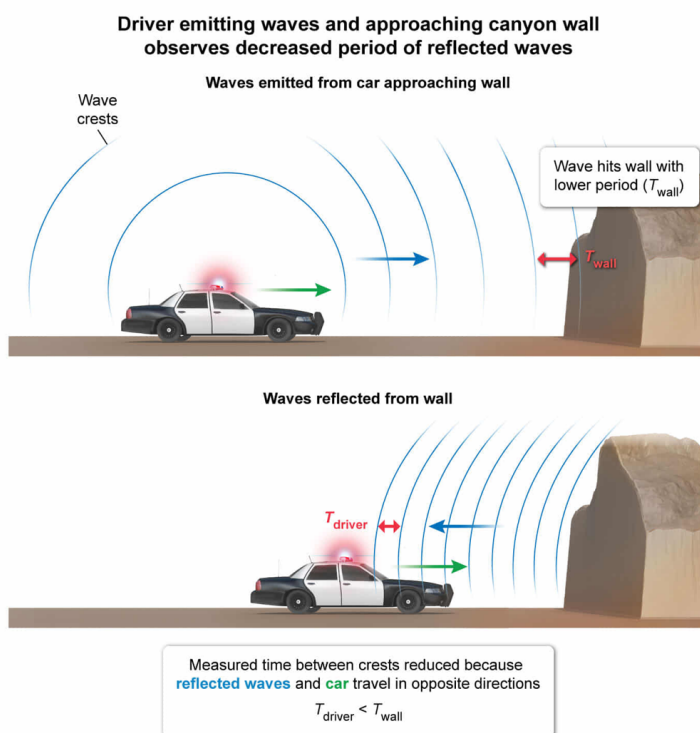
A car drives at a constant speed toward the wall of a canyon while sounding its horn at a particular frequency. The sound waves from the horn reflect from the canyon wall and return to the car as it continues moving toward the wall. What happens to the period of the reflected waves observed by the driver of the car?

- ☐ A. The period of the reflected waves will remain the same because the returning waves travel the same speed as the outgoing waves. [24%]
- ☐ B. The period of the reflected waves will remain the same because the speed of the car is constant. [15%]
- ✓ ☒ C. The period of the reflected waves will decrease because the observer moves in the opposite direction from the waves. [32%]
- ☐ D. The period of the reflected waves will decrease because the speed of the returning waves is increased. [28%]

Correct

32% answered correctly

Explanation:



According to the Doppler effect, the wave period measured by an observer depends on the relative motion of the source creating the wave (eg, sound) and the observer.

For example, when the observer and a source emitting waves of period T_0 move *toward* each other, the observed wave period T_{obs} is smaller than T_0 . A smaller period is observed because the source has moved closer to the observer and the measured time between emitted wave crests is reduced. Conversely, when the source and the observer move *away* from each other, T_{obs} is greater than T_0 because the waves emitted from the source travel a longer distance to the observer.

In this question, the car emits sound waves with a frequency $f_0 = 1/T_0$ as it approaches the canyon wall. The Doppler effect implies that the sound waves hit the wall with a period T_{wall} less than T_0 :

$$T_{\text{wall}} < T_0$$

The sound waves are then reflected from the wall toward the car. Consequently, the wall acts as the source and the driver as the observer. Because the driver is moving toward the wall (ie, in the opposite direction from the waves), the observed period T_{driver} is further reduced:

$$T_{\text{driver}} < T_{\text{wall}} < T_0$$

Therefore, the most accurate description of the period is Choice C.

(Choice A) Emitted or reflected sound waves travel in air at the same speed, but the relative motion of the source and observer yields a smaller period.

(Choice B) The speed of the car is constant, but because the car is moving relative to the wave, the driver observes a smaller period.

(Choice D) The period of the reflected waves decreases due to the Doppler effect, not because the wave speed increases.

Educational objective:

The period of an observed wave changes based on the relative motion between the source and the observer. When source and observer move toward each other, wave period is decreased.

Time spent:0 sec

Subject:
Physics

QID:404147

System:
Light and Sound